

Tianyi Xu

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Education

University of Wisconsin-Madison

Sept 2021 – Dec 2025

BS in Computer Science, Data Science, Mathematics

- GPA: 3.918/4.000
- **Coursework:** Foundation Models, Deep Learning in Computer Vision, Learning Based Image Synthesis and Manipulation, Data-Centric ML, Real Analysis, Probability Theory, Theoretical Data Science

Research Experiences

Research Assistant

Madison, WI

Advisor: Junjie Hu

July 2025 – Present

- Developed **tone-aware, speaker-invariant** speech representations for **low-resource tonal languages** by adapting multilingual self-supervised speech encoders (wav2vec2/ XLS-R) using fairseq.
- Built a **multitask training** pipeline with contrastive learning and ASR CTC auxiliary loss to reduce speaker bias while preserving tonal distinctions, regularized with **knowledge distillation**.
- Benchmarked evaluation via **cross-speaker retrieval**, achieving **0.95-0.99** Top-1/Top-5 accuracy consistently while maintaining good tone distinction. Manuscript in preparation for ACL 2026.

Research Intern, Epic Lab (SJTU)

Remote

Advisor: Linfeng Zhang, Mentor: Shaobo Wang

Aug 2025 – Present

- Built a **multi-GPU** GPT-2 pretraining pipeline (distributed training) and implemented Llama3/Qwen3 pretraining with **Megatron** framework.
- Developed **online, validation-guided data selection** and **domain-mix adaptation** modules, and integrated them with optimizers such as **Adam** and **Muon** for stable long-run training.
- Ran ablation studies on data mixes/selection cadences; evaluated on **MMLU**, **HellaSwag**, **OpenBookQA**, etc; improving accuracy for **+2–3 points** on average. Manuscript in preparation for ICML26.

Research Assistant

Madison, WI

Advisor: Claudia Solís-Lemus

Jan 2023 – Present

- Analyzed **large-scale, high-dimensional microbiome datasets** using **visualization, dimensionality reduction, regression/statistical modeling**, and **network analysis** in **R** to identify ecological and microbial association patterns.
- Developed a **label-efficient bioacoustic detection** pipeline for biodiversity monitoring: pretrained an **AudioMAE** encoder on unlabeled rainforest soundscapes and integrated it into a detector with **self-training**, improving **~0.3 mAP** over supervised baselines under out-of-distribution site/time shifts.

Independent Research Study

Madison, WI

Advisor: Pedro Morgado

Jan 2025 – May 2025

- Built a **latent generative pipeline** for vision-guided audio separation (VAE latent space) that cuts compute/memory while preserving quality.
- Implemented and compared **latent diffusion** and **rectified flow matching** with visual conditioning; introduced **per-channel z-norm + clipping** that stabilized training.
- Built a visual-guided audio diffusion pipeline with U-Net conditioning; evaluated with FAD, LPAPS, CLAP, and ImageBind on AVE/Music datasets.

Publications

SITA: Learning Speaker-Invariant and Tone-Aware Speech Representations for Low-Resource Tonal Languages

Under Review

Tianyi Xu*, Xuan Ouyang*, Binwei Yao, Shoua Xiong, Sara Misurelli, Maichou Lor, Junjie Hu

Submitted to ACL 2026.

OPUS: Towards Efficient and Principled Data Selection in Large Language Model Pre-training in Every Iteration <i>(Blog)</i> ↗	Under Review
Shaobo Wang*, Xuan Ouyang*, Tianyi Xu* , Jialin Liu, Guo Chen, Tianyu Zhang, Yuzheng Hu, Junhao Zheng, Yubo Ma, Kexin Yang, Xingzhang Ren, Dayiheng Liu, Linfeng Zhang	
<i>Submitting to ICML 2026</i>	
Self-Supervised Sound Detection with AudioMAE for Robust, Label-Efficient Biodiversity Monitoring	Under review
Tianyi Xu , Claudia Solís-Lemus, Daniel Pimentel-Alarcon, Zuzana Burivalova	
<i>Under review at CV4EO, WACV 2026</i>	
<i>In preparation for Journal Submission.</i>	
Combined effects of methyl bromide and soil amendments on soil bacterial and fungal communities in turfgrass <i>(Preprint)</i> ↗	Under review
Tianyi Xu* , Salma Mukhtar*, Evan Gorstein, Claudia Solís-Lemus, Ming Yi Chou, Paul Koch	
<i>Submitted to Soil Use and Management.</i>	
Stem rot affects the structure of rhizosphere microbiome in berseem clover (Trifolium alexandrinum) (Published in Rhizosphere)	Apr 2025
Salma Mukhtar, Zain Ahmad, Noor Khan, Tianyi Xu , Dalaq Aiysha	
10.1016/j.rhisph.2025.101073 ↗	
BiomedBank: A Large-scale, Multimodal Data Ecosystem for Advancing Biomedical AI	In preparation
TBD	
<i>Manuscript in preparation.</i>	

Talks / Presentations

Friday at 4 Seminar – Invited Talk	Dec 2025
<i>“Self-Supervised Sound Detection with AudioMAE for Robust, Label-Efficient Biodiversity Monitoring”</i>	
Department of Plant Pathology, University of Wisconsin–Madison	
Undergraduate Research Symposium – Poster Presentation	Apr 2025
27th Annual Undergraduate Symposium, University of Wisconsin–Madison	
<i>“Biodiversity Detection Using Acoustic Signals with Deep Learning”</i>	
Presented as part of the requirement of Holstrom Environmental Fellowship on the progress of the project.	
Abstract Link (Page 164) ↗	
Undergraduate Research Symposium – Poster Presentation	Apr 2024
26th Annual Undergraduate Symposium, University of Wisconsin–Madison	
<i>“Biodiversity Detection Using Acoustic Signals with Deep Learning”</i>	
Abstract Link (Page 139) ↗	

Awards

Phi Beta Kappa (PBK), Inducted	2025
Holstrom Environmental Research Fellowship	2024
Awarded \$3,000 + \$1,000 (advisor) for research on acoustic event detection in tropical rainforests using deep learning.	
UW–Madison Summer Scholarship	2024
Awarded \$1,000 for Summer 2024 studies in recognition of outstanding academic achievement.	
Dean’s List (Six Semesters)	2021–2024

Recognized for maintaining a GPA above 3.85 in Fall 2021, Spring 2022, Fall 2022, Fall 2023, Spring 2024, and Fall 2024.

Other Experiences & Projects

Project Assistant / Intern Remote

Nov 2024 – May 2025

- Contributed to *BiomedBank*, a large-scale multimodal biomedical dataset project, collaborated with Prof. Lichao Sun(Lehigh) and Prof. Chaowei Xiao(UW-Madison/JHU).
- Conducted literature review on data quality and built preprocessing/deduplication pipeline for biomedical NLP. Benchmarked biomedical QA and Clinical Trial Dataset with VLLM deployed medical/general Large Language Models (e.g Med-LLaMa3, GPT-OSS, Qwen3).

Project Assistant Wisconsin Alzheimer's Disease Research Center

Sept 2023 – Aug 2024

- Developed diagnosis prediction pipeline with SMOTE, Random Forest, and XGBoost.
- Explored CNN and phylogenetic embeddings for feature representation; used VAE for data synthesis.

Peer Mentor UW-Madison Department of Computer Science

Sept 2022 – May 2024

- Mentored 400+ students in CS 220, 320, and 540 on Python, AI, and data science topics.
- Hosted 400+ office hours covering OOP, visualization, and ML/DL fundamentals.

Developer UW-Madison DoIT Helpdesk

Oct 2021 – Aug 2022

- Improved UI/UX with dark mode enhancements and user preference handling.
- Built and documented APIs integrated with MySQL for efficient data access.
- Created staff filtering system in React.js to support searches across 1000+ employees.

Selected Graduate Course Projects/Papers University of Wisconsin-Madison

2023-2024

- *ABTUNE – Alpha Batch Size Tuning (Graduate Foundation Model Course, Full Mark)*: Benchmarked two-phase training and adaptive batch size strategies on noisy vs. clean data across image and text classification.
- *Learning Smartly – Data-Centric Methods for LMs (Graduate-Level Data-Centric ML Course, Full Mark)*: Developed a hybrid framework combining LLM-generated label functions and active learning to improve weak supervision in text classification tasks.
- *Image Style Transfer with Diffusion Models (STAT 453 Deep Learning, A)*: Supervised fine-tuned (SFT) Stable Diffusion using curated paired datasets (CycleGAN, AdaIN, GPT-4 prompts) for semantic-preserving, text-guided multi-style transfer.
- *ImageEditNet – SFT on InstructPix2Pix (Graduate Learning-Based CV Course, A)*: Improved real-image text-guided editing via SFT on MagicBrush and GAN-generated datasets, boosting CLIP similarity and facial editing quality.
- *ASL Fingerspelling Recognition (CS 539 Neural Network, Full Mark)*: Fine-tuned Faster R-CNN (ResNet50, MobileNetV2) for real-time ASL letter detection, achieving 0.913 IoU and 0.7 accuracy; deployed webcam demo for live inference.

Skills

Programming Python, R, C/C++, Java, SQL, JavaScript, HTML/CSS

ML/DL: PyTorch, TensorFlow, Hugging Face (Transformers, Diffusers), PyTorch Geometric, Scikit-Learn

LLM Training / Systems: Megatron(-LM), distributed training (multi-GPU), CUDA

Data / Infra: Pandas, Spark, HDFS, BigQuery, Cassandra, Kafka

Tools: Git, Linux, Docker, GCP, MySQL, SQLite, Flask, Selenium, Streamlit, Snowflake, Elasticsearch, Airbyte, Weights & Biases

Languages: Chinese (Mandarin, native), English (fluent), Japanese (basic)